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DEPARTMENT OF COMPUTER ENGINEERING

Laboratory Manual

Class: B.E. Computer

Subject: Cyber Security Lab

Academic Year: 2025-26

Semester: VIII

DEPARTMENT OF COMPUTER ENGINEERING

Vision of the Department

To emerge as the leading Computer Engineering department for inclusive development of students.

Mission of the Department

To provide student-centered conducive environment for preparing knowledgeable, competent and value-added computer engineers.

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Program Educational Objectives (PEOs)

PEO 1. Core Knowledge

Computer engineering graduates will have the knowledge of basic science and Engineering skills, Humanities, social science, management and conceptual and practical understanding of core computer engineering area with project development.

PEO 2. Employment

Computer engineering graduates will have the knowledge of Industry-based technical skills to succeed in entry level engineering position at various industries as well as in academics.

PEO 3. Professional Competency

Computer engineering graduates will have the ability to communicate effectively in English, to accumulate and disseminate the knowledge and to work effectively in a team with a sense of social awareness.

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Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

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Program Specific Outcomes (PSOs)

Computer Engineering Graduates will be able to:

1. **Software Systems Development:** Apply the theoretical concepts of computer engineering and practical knowledge in analysis, design and development of software systems.
2. **Open Source Software:** Demonstrate familiarity and practical competence with a broad range of programming languages and open source platforms.
3. **Computer Proficiency:** Exhibit proficiency through latest technologies in demonstrating the ability for work efficacy to the industry & society.

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Course Objectives for Cyber Security Lab

1. To learn Information Technology Act of India
2. To understand the importance of Cyber Security
3. To learn Offensive Cyber Security Tools
4. To learn Defensive Cyber Security Tools
5. To learn Security Testing Tools for Web Applications

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Course Outcomes (CO) for Cyber Security Lab

1. To describe Information Technology Act of India
2. Describe Cyber Security
3. Demonstrate Offensive Cyber Security Tools
4. Demonstrate Defensive Cyber Security Tools
5. Demonstrate Security Testing Tools for Web Applications

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Cyber Security Lab

Mapping of CO to PO

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
To describe Information Technology Act of India		2	3	3		3	3	3				3
Describe Cyber Security	3	2	3	3	3	3	3	3				3
Demonstrate Offensive Cyber Security Tools	3	2	3	3	3	3	3	3				3
Demonstrate Defensive Cyber Security Tools	1	1	1	1		3	3	3				3
Demonstrate Security Testing Tools for Web Applications	3	2	3	3	3	3	3	3				3
Average	2.5	1.8	2.6	2.6	3	3	3	3				3

Mapping of CO to PSO

CO	PSO 1	PSO 2	PSO 3
To describe Information Technology Act of India	3		
Describe Cyber Security	3	3	3
Demonstrate Offensive Cyber Security Tools	3	3	3
Demonstrate Defensive Cyber Security Tools	3	2	2
Demonstrate Security Testing Tools for Web Applications	3	3	3
Average	3	2.75	2.75

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Cyber Security Lab

List of Experiments

Experiment No.	Title of Experiment
1	Study of Information Technology Act–Indian Perspective
2	Study of recent cyber incidents/ vulnerability
3	Study of information gathering tools in Kali Linux
4	Study of vulnerability analysis tools in Kali Linux
5	Study of web application analysis tools in Kali Linux
6	Study of database assessment tools in Kali Linux
7	Study of sniffing and spoofing tools in Kali Linux
8	Study of forensics tools in Kali Linux
9	Study of reporting tools in Kali Linux